

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> Carla is downloading a 200 GB file. So, normally she can download 2 GB / minute. This means, $200 \text{ GB} / 2 \text{ GB / minute} = 100 \text{ minutes}$ would be taken to download the file normally. But, 40% of the way through the download, Windows forces a restart to install updates. So, she has downloaded 40% of 200 GB which is $(40 / 100) * 200 = 80 \text{ GB}$. Time taken to download 80 GB is $80 \text{ GB} / 2 \text{ GB / minute} = 40 \text{ minutes}$. Then, Carla has to restart the download from the beginning. So, total time taken so far is 40 minutes. After restart, she has to download 200 GB again. So, time taken to download 200 GB is $200 \text{ GB} / 2 \text{ GB / minute} = 100 \text{ minutes}$. Adding the time taken after restart, total time taken is $40 + 100 = 140 \text{ minutes}$. So, total time taken to download the file is 140 minutes</think><answer>140</answer>

